

OE-EC704B — Optimization Technique

Semester VII • 3L:0T:0P • 3 Credits

Module 3 Exam Notes: Gradient-Based Methods

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

Contents

1	Fundamentals of Gradient-Based Search	2
2	Newton-Raphson Method	2
2.1	Derivation of the Iterative Formula	2
2.2	Convergence and Characteristics	2
3	Bisection Method	3
3.0.1	Algorithm Steps	3
4	Secant Method	3
4.1	Iterative Formula	4
5	Computer Programs / Pseudo-Codes	4

1. Fundamentals of Gradient-Based Search

Gradient-based single-variable methods use derivative information to find stationary points by solving:

$$g(x) = f'(x) = 0 \quad (1)$$

These algorithms are highly efficient because derivatives provide the slope and curvature of the objective function, indicating the direction and distance to the optimum.

2. Newton-Raphson Method

The Newton-Raphson method is a second-order method that approximates the function locally with a quadratic Taylor series expansion.

2.1 Derivation of the Iterative Formula

Let x_k be the current estimate of the optimum. Expanding the first derivative $f'(x)$ about x_k using a first-order Taylor series yields:

$$f'(x) \approx f'(x_k) + f''(x_k)(x - x_k)$$

To locate the stationary point (x_{k+1} where $f'(x_{k+1}) = 0$):

$$f'(x_k) + f''(x_k)(x_{k+1} - x_k) = 0$$

Rearranging for x_{k+1} gives the standard **Newton-Raphson Update**:

Newton-Raphson Iterative Equation

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)} \quad (2)$$

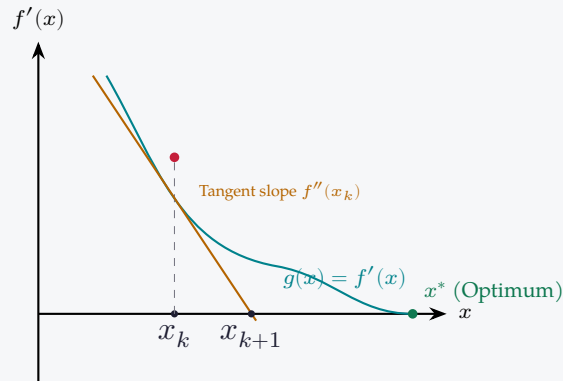
Where:

- $f'(x_k)$ is the first derivative (slope) at step k .
- $f''(x_k)$ is the second derivative (curvature) at step k .

2.2 Convergence and Characteristics

- **Quadratic Convergence:** Near the optimum, the error decreases quadratically ($e_{k+1} \propto e_k^2$), making it the fastest gradient-based method.
- **Limitations:** Requires computing the analytical second derivative $f''(x)$. If $f''(x_k) = 0$, the method fails. It is highly sensitive to the initial guess x_0 and can diverge if starting far from the root.

Graphical projection of Newton-Raphson



3. Bisection Method

The bisection method is a robust first-order bracketing technique that solves $f'(x) = 0$ by halving intervals based on sign changes.

Bisection Criterion

Let the initial interval be $[a, b]$. If $f'(x)$ is continuous and:

$$f'(a) \cdot f'(b) < 0 \quad (3)$$

then by the Intermediate Value Theorem, at least one stationary point x^* exists in $[a, b]$ where $f'(x^*) = 0$.

3.0.1 Algorithm Steps

1. Calculate the midpoint: $x_0 = \frac{a+b}{2}$ and evaluate $f'(x_0)$.
2. If $|f'(x_0)| \leq \epsilon$ (tolerance), stop; $x^* \approx x_0$.
3. If $f'(a) \cdot f'(x_0) < 0$, the root lies in $[a, x_0]$. Set $b = x_0$.
4. If $f'(x_0) \cdot f'(b) < 0$, the root lies in $[x_0, b]$. Set $a = x_0$.
5. Repeat until $|b - a| \leq \epsilon$. Convergence is guaranteed, though linear and slow.

4. Secant Method

The secant method approximates the second derivative $f''(x)$ in the Newton-Raphson formula using two previous evaluations, avoiding analytical differentiation.

4.1 Iterative Formula

The curvature is approximated using finite differences between x_k and x_{k-1} :

$$f''(x_k) \approx \frac{f'(x_k) - f'(x_{k-1})}{x_k - x_{k-1}} \quad (4)$$

Substituting this into the Newton equation yields the **Secant Iterative Update**:

Secant Method Equation

$$x_{k+1} = x_k - f'(x_k) \left[\frac{x_k - x_{k-1}}{f'(x_k) - f'(x_{k-1})} \right] \quad (5)$$

This method requires two initial guess points x_0 and x_1 , and converges superlinearly with a convergence rate of:

$$\text{Rate} \approx 1.618 \quad (6)$$

5. Computer Programs / Pseudo-Codes

Newton-Raphson Optimization Pseudo-code

Input: Initial guess x_0 , tolerance ϵ , max_iter

Output: Local minimum coordinate x^*

```
Function NewtonRaphson(x0, eps, max_iter):
    x = x0
    For k = 1 to max_iter:
        df = FirstDerivative(x)
        ddf = SecondDerivative(x)
        If |ddf| < 1e-12:
            Print "Error: Curvature close to zero!"
            Return null
        x_new = x - df / ddf
        If |x_new - x| < eps or |df| < eps:
            Return x_new
        x = x_new
    Return x
```

Secant Optimization Pseudo-code

Input: Two initial guesses x_0 , x_1 , tolerance ϵ , max_iter

Output: Local minimum coordinate x^*

```
Function SecantMethod(x0, x1, eps, max_iter):
    x_prev = x0
```

```
x_curr = x1
For k = 1 to max_iter:
    df_prev = FirstDerivative(x_prev)
    df_curr = FirstDerivative(x_curr)
    If |df_curr - df_prev| < 1e-12:
        Print "Error: Denominator is zero!"
        Return null
    x_next = x_curr - df_curr * (x_curr - x_prev) / (df_curr - df_prev)
    If |x_next - x_curr| < eps:
        Return x_next
    x_prev = x_curr
    x_curr = x_next
Return x_curr
```

Quick Revision — Module 3

Gradient-Based Methods | OE-EC704B

Key Formulas & Convergence Rates

Newton-Raphson Formula	$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}.$
Newton Convergence	Quadratic rate: $e_{k+1} \approx Ce_k^2.$
Bisection Condition	$f'(a) \cdot f'(b) < 0$ (signs must be opposite).
Secant Update Formula	$x_{k+1} = x_k - f'(x_k) \left[\frac{x_k - x_{k-1}}{f'(x_k) - f'(x_{k-1})} \right].$
Secant Convergence	Superlinear rate: $e_{k+1} \approx Ce_k^{1.618}.$

One-Line Definitions

- **Gradient-Based Optimization:** Minimizing/maximizing functions using directional slope (derivatives) rather than search comparisons alone.
- **Newton-Raphson Method:** An iterative root-finding technique using tangent lines to estimate the point where the derivative is zero.
- **Bisection Method:** A root-finding algorithm that repeatedly bisects an interval and selects a sub-interval in which a sign change occurs.
- **Secant Method:** A gradient-based optimization method that replaces analytical second derivatives with finite difference slope approximations.
- **Quadratic Convergence:** A convergence behavior where the number of correct decimal places roughly doubles with each iteration.
- **Intermediate Value Theorem:** Theorem stating that if a continuous function has opposite signs at interval boundaries, it must cross zero inside.

Frequently Asked / Viva Questions

Q1. Compare Newton-Raphson and Secant methods based on derivative requirements.

Newton-Raphson requires evaluating both the first derivative $f'(x)$ and the analytical second derivative $f''(x)$ at each step. The Secant method only requires the first derivative $f'(x)$ because it approximates $f''(x)$ using finite differences.

Q2. State the convergence rate of the Newton-Raphson method and define what it means.

It converges quadratically ($e_{k+1} \propto e_k^2$). This means the error in step $k + 1$ is proportional to the square of the error in step k , leading to extremely rapid convergence near the root.

Q3. Why does the Newton-Raphson method fail if $f''(x_k) = 0$?

Frequently Asked / Viva Questions (contd.)

Because the formula involves dividing by the second derivative. If $f''(x_k) = 0$, the tangent to $f'(x)$ becomes parallel to the x-axis, and the projection point x_{k+1} moves to infinity (division by zero error).

Q4. Explain the role of the sign check $f'(a) \cdot f'(b) < 0$ in the Bisection method.

It guarantees that the first derivative $f'(x)$ changes sign across the interval $[a, b]$. For a continuous derivative, this ensures that at least one stationary point (root of $f'(x) = 0$) lies within the interval.

Q5. What are the primary disadvantages of the Bisection method?

It has a slow, linear convergence rate. It does not utilize the magnitude of the derivative, only its sign, making it take many iterations compared to Newton's method.

Q6. Write down the Secant method update equation.

$$x_{k+1} = x_k - f'(x_k) \left[\frac{x_k - x_{k-1}}{f'(x_k) - f'(x_{k-1})} \right].$$

Q7. How does the Secant method differ from the regular False Position (Regula Falsi) root-finding method?

False Position always brackets the root and retains points of opposite signs. The Secant method simply uses the last two calculated points, meaning it is not guaranteed to bracket the root and can occasionally diverge.

Q8. What is the convergence rate of the Secant method?

It converges superlinearly with order $\alpha = \frac{1+\sqrt{5}}{2} \approx 1.618$ (the Golden Ratio).

Q9. Under what conditions can the Newton-Raphson method diverge?

It can diverge if the initial guess x_0 is chosen far from the optimum, especially in regions with inflection points, local extrema of $f'(x)$, or flat regions where $f''(x) \approx 0$.

Q10. Write a short MATLAB/Python routine snippet for updating the Secant point.

```
x_next = x_curr - df_curr * (x_curr - x_prev) / (df_curr - df_prev)
where df is the derivative function.
```